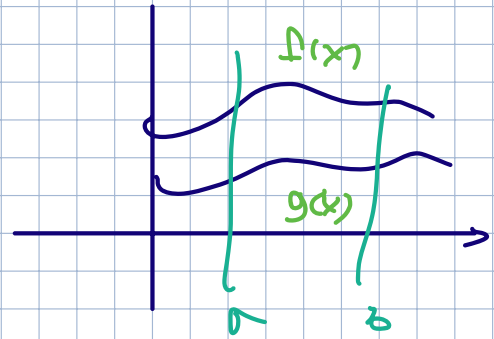


:= 0, 0, 0

$$\int_a^b f(x) - g(x) dx$$



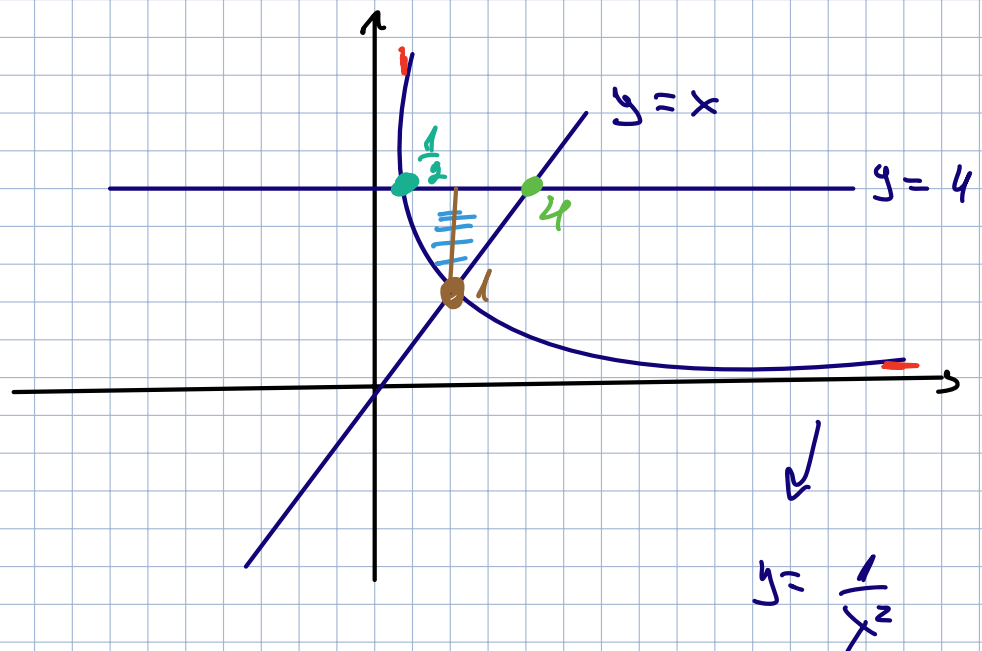
$$y = x$$

$$y = 4$$

$$y = \frac{1}{x^2}$$

:= 0, 0, 0

re 200



$$4 = \frac{1}{x^2}$$

$$4x^2 = 1$$

$$x^2 = \frac{1}{4}$$

$$x = \frac{1}{2}$$

$$y = \frac{1}{x^2}$$

$$x = \frac{1}{x^2}$$

$$x^3 = 1$$

$$x = 1$$

$$S = \int_{1/2}^1 \left(4 - \frac{1}{x^2}\right) dx + \int_1^4 (4 - x) dx$$

$$= 4x + \frac{1}{x} \Big|_{1/2}^1 + 4x - \frac{x^2}{2} \Big|_1^4 =$$

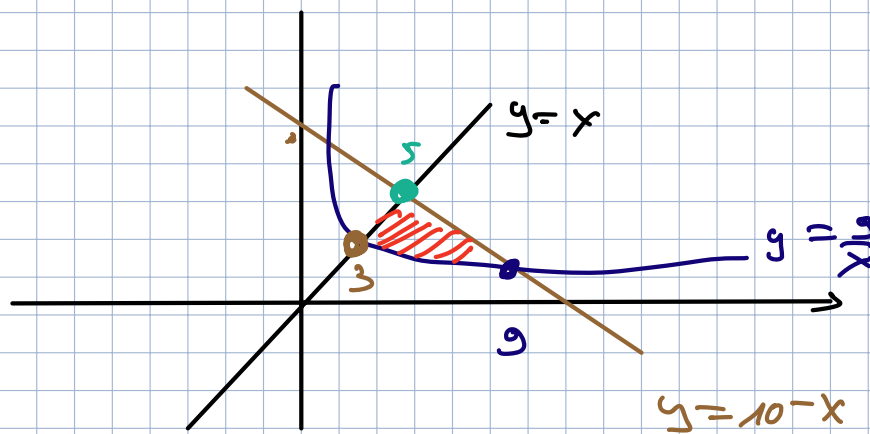
$$\left(4 \cdot 1 + \frac{1}{1}\right) - \left(4 \cdot \frac{1}{2} + \frac{1}{1/2}\right) + \left(4 \cdot 4 - \frac{4^2}{2}\right) - \left(4 \cdot 1 - \frac{1^2}{2}\right)$$

$$xy = 9 \rightarrow y = \frac{9}{x}$$

$$x + y = 10 \rightarrow y = 10 - x$$

$$y = x$$

$$y \leq x \quad \text{verkt}$$



$$x = \frac{g}{x}$$

$$x^2 = g$$

$$x = 3$$

$$x = 10 - x$$

$$2x = 10$$

$$x = 5$$

$$\frac{g}{x} = 10 - x$$

$$g = 10x - x^2$$

$$x^2 - 10x + g = 0$$

$$x = 9$$

$$\int_3^5 x - \frac{g}{x} dx + \int_5^9 10 - x - \frac{g}{x} dx$$

$$= \left[\frac{x^2}{2} - g \ln|x| \right]_3^5 + \left[10x - \frac{x^2}{2} - \ln|x| \right]_5^9$$

$$y = \ln^2 x$$

$$y = \ln x$$

∴ man

$$\ln^2 x = \ln x$$

2.1.1

$$\ln^2 x - \ln x = 0$$

$$\ln x (\ln x - 1) = 0$$

$$\swarrow$$

$$\ln x = 0$$

$$x = e^0$$

$$x = 1$$

$$\downarrow$$

$$\ln x - 1 = 0$$

$$\ln x = 1$$

$$x = e$$

$$y_1 = \ln x \quad |_{x=2} = 0.2$$

$$y_2 = \ln^2 x \quad |_{x=2} = 0.5$$



$$\int_1^e \ln x - \ln^2 x \, dx$$

$$\int \ln x \, dx = \left| \begin{array}{ll} u = \ln x & du = \frac{1}{x} dx \\ dv = dx & v = x \end{array} \right|$$

$$= x \ln x - \int x \cdot \frac{1}{x} dx = x \ln x - x$$

$$\int \ln^2 x \, dx = \left| \begin{array}{l} u = \ln^2 x \rightarrow du = 2 \ln x \cdot \frac{1}{x} \\ dv = dx \rightarrow v = x \end{array} \right.$$

$$= x \cdot \ln^2 x - \int x \cdot 2 \ln x \cdot \frac{1}{x} dx$$

$$= x \ln^2 x - 2 \int \ln x \, dx$$

$$x \ln^2 x - 2 [x \ln x - x]$$

$$\left(x \ln x - x - [x \ln^2 x - 2(x \ln x - x)] \right) \begin{matrix} \uparrow \\ \leftarrow \\ \leftarrow \\ \downarrow \end{matrix}$$

$$y^a = ax + 1$$

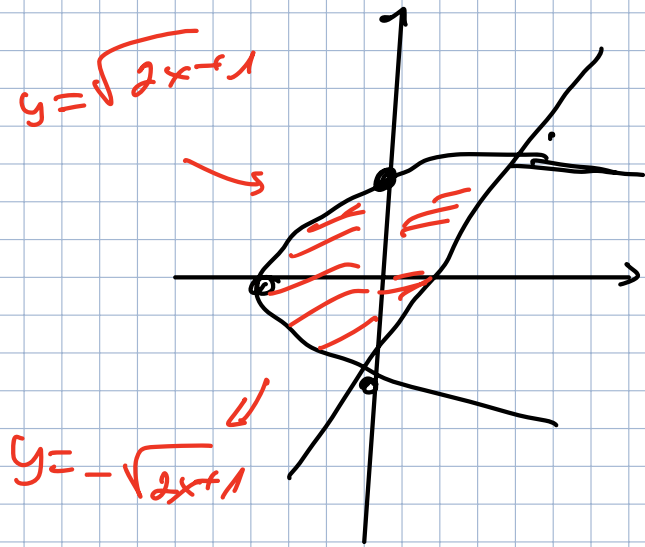
$$y = x - 1$$

$$n \in \mathbb{N} \quad \frac{1}{n} > 0$$

$$: \mathbb{R} \setminus \mathbb{Q}$$

$$\frac{y^2 - 1}{2} = x$$

$$x = y + 1$$



$$\frac{y^2 - 1}{2} = y + 1$$

$$y^2 - 1 = 2y + 2$$

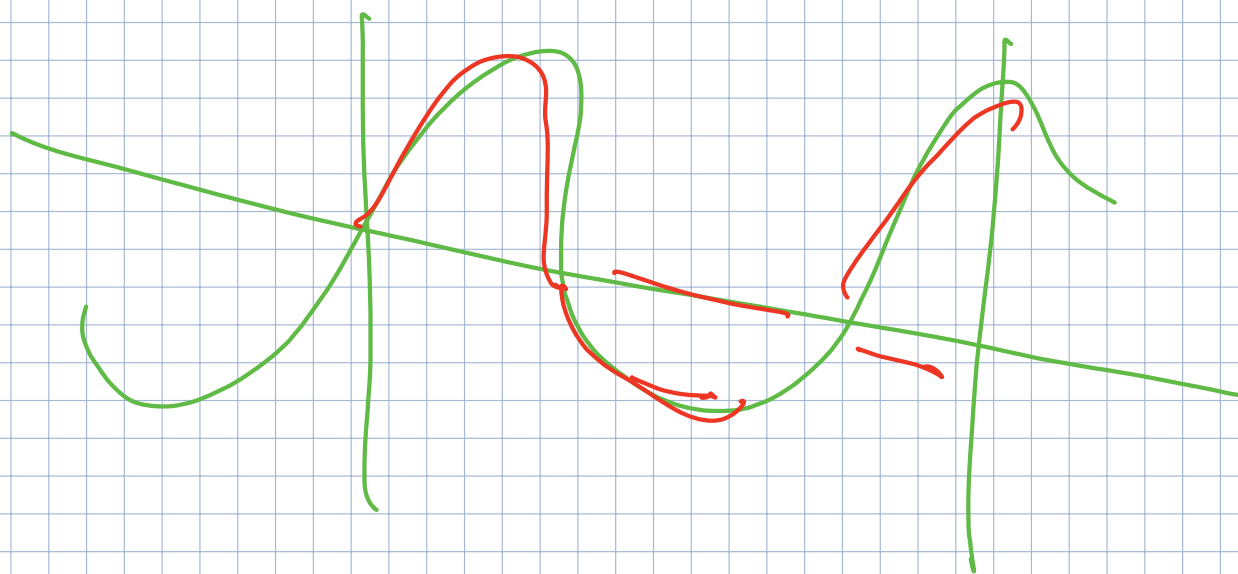
$$y^2 - 2y - 3 = 0$$

$$y = 3 \quad y = -1$$



$$\int_{-1}^3 y + 1 - \left(\frac{y^2 - 1}{2} \right) dy$$

$$= \frac{y^2}{2} + y - \frac{y^3}{2 \cdot 3} + \frac{1}{2} y \Big|_{-1}^3$$



הערה

$$\int_a^b \sqrt{1 + (f'(x))^2} \, dx$$

הערה

:- wir gehen vor

$$f(x) = \frac{x^3}{6} + \frac{1}{2x}$$

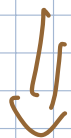
$$1 \leq x \leq 2$$

$$f'(x) = \frac{3x^2}{6} - \frac{1}{2x^2} = \frac{x^2}{2} - \frac{1}{2x^2}$$

$$\int_1^2 \sqrt{1 + \left(\frac{x^2}{2} - \frac{1}{2x^2} \right)^2} dx$$

$$\int_1^2 \sqrt{1 + \frac{x^4}{4} - 2 \cdot \frac{x^2}{2} \cdot \frac{1}{2x^2} + \frac{1}{4x^4}} dx$$

$$\int_1^2 \sqrt{\frac{1}{2} + \frac{x^4}{4} + \frac{1}{4x^4}} dx$$



$$\int_1^2 \sqrt{\left(\frac{x^2}{2} + \frac{1}{2x^2}\right)^2} dx$$

$$\int_1^2 \left(\frac{x^2}{2} + \frac{1}{2x^2}\right) dx$$

$$= \left. \frac{x^3}{2 \cdot 3} + \frac{1}{2x} \right|_1^2$$

701

$$f(x) = \sqrt{(x+1)^3}$$

701k
= 701

$$[-1, 4] \quad \text{867} \quad \text{807}$$

$$f(x) = (x+1)^{\frac{3}{2}}$$

$$f'(x) = \frac{3}{2} (x+1)^{1/2}$$

$$\int_{-1}^4 \sqrt{1 + \left(\frac{3}{2} (x+1)^{1/2} \right)^2}$$

$$\int_{-1}^4 \sqrt{1 + \frac{9}{4} (x+1)} \, dx$$

$$\int_{-1}^4 \sqrt{\frac{9}{4}x + \frac{13}{4}} dx$$

$$\frac{\left(\frac{9}{4}x + \frac{13}{4}\right)^{3/2}}{3/2 \cdot \frac{9}{4}} \Bigg|_{-1}^4$$

הצגה

y ציר

x ציר

$$y = f(x) \quad a \leq x \leq b$$

(y=0)

$$y = f(x) \quad a \leq x \leq b$$

(y=0)

הצגה

הצגה

$$V = \int_a^b x f(x) dx$$

$$V = \int_a^b f(x) dx$$

$$x = f(y) \quad c \leq y \leq d$$

(x=0)

$$x = f(y) \quad c \leq y \leq d$$

(x=0)

הצגה

הצגה

$$V = \int_c^d f^2(y) dy$$

$$V = \int_c^d y f(y) dy$$

הצגה

$$y = e^{x^2}$$

$$y = 0$$

$$x = 1$$

$$x = 0$$

y ציר

הצגה

$$V = 2\pi \int_0^1 x \cdot e^{x^2} dx$$

$$t = x^2$$

$$dt = 2x dx$$

$$x=0 \rightarrow t=0^2=0$$

$$x=1 \rightarrow t=1^2=1$$

$$= \pi \int_0^1 e^t dt = \pi \cdot e^t \Big|_0^1 =$$

$$\pi (e^1 - e^0)$$

השאלה היא כיצד למצוא את האינטגרל

$$\begin{aligned} y &= x^2 \\ y^2 &= x \end{aligned} \quad \rightarrow \quad y = \sqrt{x}$$

התשובה היא כיצד למצוא את האינטגרל

$$y = x^2 \rightarrow y^2 = x^4$$

$$y^2 = x$$

$$x^4 = x$$

$$x(x^3 - 1) = 0$$

$$x = 0$$

$$x = 1$$

$$V = \pi \int_a^b f^2(x) dx$$

$$V = \pi \int_a^b f^2(x) - g^2(x) dx$$

$$f = x^2$$

$$g = \sqrt{x}$$

$$= \pi \int_0^1 (\sqrt{x})^2 - (x^2)^2 dx$$

$$\begin{array}{c} | \quad | \quad | \\ 0 \quad \frac{1}{2} \quad 1 \end{array}$$

$$\pi \int_0^1 x - x^4 dx =$$

$$y_1\left(\frac{1}{2}\right) = \frac{1}{4}$$

$$y_2\left(\frac{1}{2}\right) = \sqrt{\frac{1}{2}} = 0.7$$

$$\pi \left(\frac{x^2}{2} - \frac{x^5}{5} \right) \Big|_0^1$$

$$\pi \cdot \left(\frac{1}{2} - \frac{1}{5} \right) = \pi \frac{(5-2)}{10} = \frac{\pi \cdot 3}{10}$$

המשטח הנמצא מתחת קו העקום

ממשטח $x=1$ ו- $y=2x-x^2$

$y=0$ ו- $x=3$

$$V = 2\pi \int_a^b x f(x) dx$$

$$0 = 2x - x^2$$

$$0 = x(2-x)$$

$$\begin{array}{cc} \downarrow & \searrow \\ x=0 & x=2 \end{array}$$

$$V = 2\pi \int_0^2 x(2x - x^2) dx$$

לגובה y וזווית x

$$0 \leq x < \frac{\pi}{2}$$

$$y = \tan x$$

$$y = \sin x$$

לגובה y וזווית x

$$\tan x = \sin x$$

$$\frac{\sin x}{\cos x} = 2 \sin x$$

$$\sin x \left(\frac{1}{\cos x} - 2 \right) = 0$$

$$\sin x = 0$$

$$x = \pi k$$

$$\downarrow$$

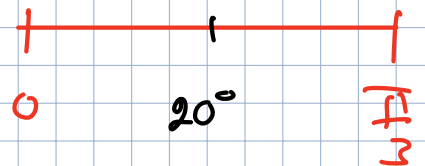
$$x = 0$$

$$\frac{1}{\cos x} = 2$$

$$\cos x = \frac{1}{2}$$

$$x = \frac{\pi}{3}$$

$$V = \frac{1}{H} \int_a^b f(x) dx$$



$$y_1 = \tan x$$

$$y_2 = 2 \sin x$$

$$y_1 = \tan(20^\circ) = 0.36$$

$$y_2 = 2 \sin(20^\circ) = 0.68$$

$$V = \pi \int_0^{\frac{\pi}{2}} (2 \sin x)^2 - (\tan x)^2 dx$$

$$4 \int \sin^2 x dx = 4 \int \frac{1 - \cos(2x)}{2} dx$$

$$= 4 \left[\frac{1}{2} x - \frac{\sin(2x)}{2 \cdot 2} \right]$$

$$\int \tan^2 x = \int \frac{\sin^2 x}{\cos^2 x} dx = \int \frac{1 - \cos^2 x}{\cos^2 x}$$

$$= \int \frac{1}{\cos^2 x} - 1 dx$$

$$= \tan x - x$$

$$\pi \left[4 \left[\frac{1}{2}x - \frac{\sin(2x)}{2 \cdot 2} \right] - (\tan x - x) \right]_0^{\frac{\pi}{2}}$$

המשפט של ברנולי

$$y = 4 - x^2$$

$$y = 3x$$

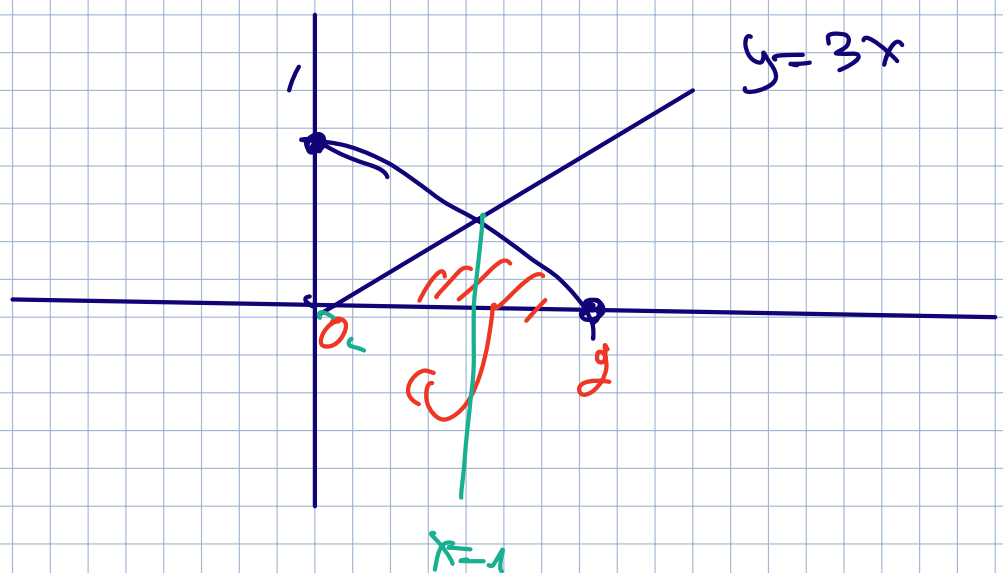
המשפט של ברנולי $[0, 2]$ x, y

המשפט של ברנולי x, y $[0, 2]$

$$4 - x^2 = 3x$$

$$x^2 + 3x - 4 = 0$$

$$x = 1, \text{ } \cancel{x = 4}$$



$$V = \pi \int_a^b f(x)^2 dx$$

$$V = \pi \int_0^1 (3x)^2 dx + \pi \int_1^2 (4-x^2)^2 dx$$

$$\pi \int_0^1 9x^2 dx + \pi \int_1^2 (16 - 8x^2 + x^4) dx$$

$$\pi \left[\frac{9x^3}{3} \right]_0^1 + \pi \left(16x - \frac{8x^3}{3} + \frac{x^5}{5} \right) \Big|_1^2$$